JR Validated Environment — Acceptance Sampling Module

OQ Validation Plan — JR-VP-AS-001 v1.0

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1. Purpose and Scope

This Operational Qualification (OQ) Validation Plan defines the test cases for the Acceptance Sampling (AS) module of the JR Validated Environment. The AS module comprises four R scripts: jrc_as_attributes, jrc_as_variables, jrc_as_oc_curve, and jrc_as_evaluate. These scripts are used for designing acceptance sampling plans and applying them to production lots.

This plan covers 44 automated test cases executed using the pytest framework via the admin_as_oq runner. Test evidence is captured in a timestamped text file in the project validation directory.

2. User Requirements

The following user requirements are tested by this plan.

UR ID	Requirement
UR-AS-001	jrc_as_attributes shall accept valid lot size, AQL, and RQL arguments and exit with code 0.
UR-AS-002	jrc_as_attributes shall produce a single sampling plan (n and c) in the output.
UR-AS-003	jrc_as_attributes shall produce a double sampling plan (n1, c1, c2) in the output when a valid plan exists.
UR-AS-004	jrc_as_attributes shall produce an OC curve table with p and Pa values.
UR-AS-005	jrc_as_attributes shall save a PNG plot to ~/Downloads/.
UR-AS-006	jrc_as_attributes shall exit non-zero and print usage when called with no arguments.
UR-AS-007	jrc_as_attributes shall exit non-zero when aql ≥ rql.
UR-AS-008	jrc_as_attributes shall exit non-zero when aql > 1 or < 0.
UR-AS-009	jrc_as_attributes shall exit non-zero when lot_size < 2.
UR-AS-010	jrc_as_attributes shall exit non-zero when --alpha is out of range.
UR-AS-011	jrc_as_attributes shall exit non-zero and report RENV_PATHS_ROOT when called directly via Rscript without the wrapper.
UR-AS-012	jrc_as_variables shall accept valid lot size, AQL, RQL, and --sides arguments and exit with code 0.
UR-AS-013	jrc_as_variables shall produce an acceptability constant k that is positive in the output.
UR-AS-014	jrc_as_variables shall produce a valid plan for --sides 2 with positive k.
UR-AS-015	jrc_as_variables shall produce an OC curve table with Pa values.
UR-AS-016	jrc_as_variables shall save a PNG plot to ~/Downloads/.
UR-AS-017	jrc_as_variables shall exit non-zero when called with no arguments.
UR-AS-018	jrc_as_variables shall exit non-zero when aql ≥ rql.
UR-AS-019	jrc_as_variables shall exit non-zero when --sides value is not 1 or 2.
UR-AS-020	jrc_as_variables shall produce output indicating a sample size reduction compared to the equivalent attributes plan.
UR-AS-021	jrc_as_variables shall exit non-zero when --alpha is out of range.
UR-AS-022	jrc_as_variables shall exit non-zero and report RENV_PATHS_ROOT when called directly via Rscript without the wrapper.
UR-AS-023	jrc_as_oc_curve shall accept valid n and c arguments and exit with code 0 with Pa values in output.
UR-AS-024	jrc_as_oc_curve shall produce Pa > 0.99 at p = 0.001 for n=32, c=1.
UR-AS-025	jrc_as_oc_curve shall produce Pa values that decrease monotonically as p increases.
UR-AS-026	jrc_as_oc_curve shall accept the --lot-size flag without error.
UR-AS-027	jrc_as_oc_curve shall save a PNG plot to ~/Downloads/.
UR-AS-028	jrc_as_oc_curve shall accept --aql and --rql flags and display AQL in output.

UR-AS-029	jrc_as_oc_curve shall exit non-zero when called with no arguments.
UR-AS-030	jrc_as_oc_curve shall exit non-zero when c ≥ n.
UR-AS-031	jrc_as_oc_curve shall exit non-zero when n ≤ 0.
UR-AS-032	jrc_as_oc_curve shall exit non-zero and report RENV_PATHS_ROOT when called directly via Rscript without the wrapper.
UR-AS-033	jrc_as_evaluate attributes mode shall produce ACCEPT verdict for a lot with defective count ≤ c.
UR-AS-034	jrc_as_evaluate attributes mode shall produce REJECT verdict for a lot with defective count > c.
UR-AS-035	jrc_as_evaluate variables mode shall produce ACCEPT verdict for a lot with Q-statistic ≥ k.
UR-AS-036	jrc_as_evaluate variables mode shall produce REJECT verdict for a lot with Q-statistic < k.
UR-AS-037	jrc_as_evaluate variables mode shall save a PNG plot to ~/Downloads/.
UR-AS-038	jrc_as_evaluate attributes mode shall save a PNG plot to ~/Downloads/.
UR-AS-039	jrc_as_evaluate shall exit non-zero when --type is missing.
UR-AS-040	jrc_as_evaluate attributes mode shall exit non-zero when --c is missing.
UR-AS-041	jrc_as_evaluate variables mode shall exit non-zero when --k is missing.
UR-AS-042	jrc_as_evaluate attributes mode shall exit non-zero and report the missing column when the data file lacks a 'result' column.
UR-AS-043	jrc_as_evaluate variables mode shall exit non-zero and report the missing column when the data file lacks a 'value' column.
UR-AS-044	jrc_as_evaluate shall exit non-zero and report RENV_PATHS_ROOT when called directly via Rscript without the wrapper.

3. Test Environment

All test cases are executed via the admin_as_oq runner using pytest with the shared OQ virtual environment at ~/.venvs/<PROJECT_ID>_oq/. The test runner records the R version, Python version, pytest version, hostname, and timestamp in the evidence file. Tests are executed from the repos/as/oq/ directory.

Item	Specification
Language	R (≥ 4.3.0 recommended)
Test framework	pytest (≥ 8.0)
Test runner	admin_as_oq
OQ venv	~/.venvs/<PROJECT_ID>_oq/ (shared with admin_oq)
Evidence output	~/.jrscript/<PROJECT_ID>/validation/as_oq_execution_<timestamp>.txt
Validation plan ID	JR-VP-AS-001

4. Test Cases — jrc_as_attributes

TC-AS-ATTR-001 Happy path — exit 0

Tests UR(s)	UR-AS-001
Command / action	jrc_as_attributes 500 0.01 0.10
Rationale	Confirms the script accepts standard valid inputs and completes successfully.
Pass criterion	Exit code = 0. Output contains 'Attributes Sampling Plan' or equivalent plan header.

TC-AS-ATTR-002 Single plan n and c present

Tests UR(s)	UR-AS-002
Command / action	jrc_as_attributes 500 0.01 0.10
Rationale	Confirms the single sampling plan outputs both sample size n and acceptance number c.
Pass criterion	Output contains 'Sample size' with a positive integer, and 'Acceptance number' with a non-negative integer.

TC-AS-ATTR-003 Double sampling plan present

Tests UR(s)	UR-AS-003

Command / action	jrc_as_attributes 500 0.01 0.10
Rationale	Confirms the double sampling plan section is present in the output.
Pass criterion	Output contains 'Double Sampling Plan' or 'Stage 1'.

TC-AS-ATTR-004 OC curve table present

Tests UR(s)	UR-AS-004
Command / action	jrc_as_attributes 500 0.01 0.10
Rationale	Confirms the OC curve table is included in the output.
Pass criterion	Output contains 'OC Curve' and 'Pa' header.

TC-AS-ATTR-005 PNG saved to ~/Downloads/

Tests UR(s)	UR-AS-005
Command / action	jrc_as_attributes 500 0.01 0.10
Rationale	Confirms a PNG file is created in ~/Downloads/ following the script run.
Pass criterion	A file matching *_jrc_as_attributes.png exists in ~/Downloads/ with mtime ≥ test start time.

TC-AS-ATTR-006 No arguments — non-zero exit

Tests UR(s)	UR-AS-006
Command / action	jrc_as_attributes (no arguments)
Rationale	Confirms the script exits non-zero and displays usage guidance when called with no arguments.
Pass criterion	Exit code ≠ 0.

TC-AS-ATTR-007 aql ≥ rql — non-zero exit

Tests UR(s)	UR-AS-007
Command / action	jrc_as_attributes 500 0.10 0.01
Rationale	Confirms the script detects and rejects inverted AQL/RQL inputs.
Pass criterion	Exit code ≠ 0.

TC-AS-ATTR-008 aql > 1 — non-zero exit

Tests UR(s)	UR-AS-008
Command / action	jrc_as_attributes 500 1.5 0.10
Rationale	Confirms the script rejects an AQL value out of the (0, 1) range.
Pass criterion	Exit code ≠ 0.

TC-AS-ATTR-009 lot_size = 1 — non-zero exit

Tests UR(s)	UR-AS-009
Command / action	jrc_as_attributes 1 0.01 0.10
Rationale	Confirms the script rejects a lot size below the minimum (2).
Pass criterion	Exit code ≠ 0.

TC-AS-ATTR-010 --alpha out of range — non-zero exit

Tests UR(s)	UR-AS-010
Command / action	jrc_as_attributes 500 0.01 0.10 --alpha 2.0
Rationale	Confirms the script rejects an --alpha value outside (0, 1).
Pass criterion	Exit code ≠ 0.

TC-AS-ATTR-011 Bypass protection

Tests UR(s)	UR-AS-011

Command / action	Rscript repos/as/R/jrc_as_attributes.R 500 0.01 0.10 (RENV_PATHS_ROOT unset)
Rationale	Confirms that running the R script directly without the validated wrapper exits non-zero and mentions RENV_PATHS_ROOT in the error output.
Pass criterion	Exit code ≠ 0. 'RENV_PATHS_ROOT' appears in combined stdout/stderr.

5. Test Cases — jrc_as_variables

TC-AS-VAR-001 Happy path — exit 0, Variables Plan in output

Tests UR(s)	UR-AS-012
Command / action	jrc_as_variables 500 0.01 0.10
Rationale	Confirms valid one-sided inputs produce exit 0 and a plan header.
Pass criterion	Exit code = 0. Output contains 'Variables Plan' or 'Variables Sampling Plan'.

TC-AS-VAR-002 k value present and positive

Tests UR(s)	UR-AS-013
Command / action	jrc_as_variables 500 0.01 0.10
Rationale	Confirms the acceptability constant k is reported and is positive.
Pass criterion	Output contains 'constant (k)' or 'k)' or 'k =''. If a numeric k is found, k > 0.

TC-AS-VAR-003 --sides 2 produces valid plan

Tests UR(s)	UR-AS-014
Command / action	jrc_as_variables 500 0.01 0.10 --sides 2
Rationale	Confirms two-sided specification produces a valid plan with exit 0 and positive k.
Pass criterion	Exit code = 0. Output contains 'Sides: 2''. If k extracted, k > 0.

TC-AS-VAR-004 OC curve table present

Tests UR(s)	UR-AS-015
Command / action	jrc_as_variables 500 0.01 0.10
Rationale	Confirms the OC curve table is present in the output.
Pass criterion	Output contains 'OC Curve' and 'Pa''.

TC-AS-VAR-005 PNG saved to ~/Downloads/

Tests UR(s)	UR-AS-016
Command / action	jrc_as_variables 500 0.01 0.10
Rationale	Confirms a PNG file is created in ~/Downloads/ following the script run.
Pass criterion	A file matching *_jrc_as_variables.png exists in ~/Downloads/ with mtime ≥ test start time.

TC-AS-VAR-006 No arguments — non-zero exit

Tests UR(s)	UR-AS-017
Command / action	jrc_as_variables (no arguments)
Rationale	Confirms the script exits non-zero when called with no arguments.
Pass criterion	Exit code ≠ 0.

TC-AS-VAR-007 aql ≥ rql — non-zero exit

Tests UR(s)	UR-AS-018
Command / action	jrc_as_variables 500 0.10 0.01
Rationale	Confirms the script rejects inverted AQL/RQL.
Pass criterion	Exit code ≠ 0.

TC-AS-VAR-008 --sides invalid — non-zero exit

Tests UR(s)	UR-AS-019
Command / action	jrc_as_variables 500 0.01 0.10 --sides 3
Rationale	Confirms the script rejects --sides values other than 1 or 2.
Pass criterion	Exit code ≠ 0.

TC-AS-VAR-009 Sample reduction noted in output

Tests UR(s)	UR-AS-020
Command / action	jrc_as_variables 500 0.01 0.10
Rationale	Confirms the output includes a comparison with the equivalent attributes plan.
Pass criterion	Output contains 'Sample reduction', 'fewer', or 'Comparison'.

TC-AS-VAR-010 --alpha out of range — non-zero exit

Tests UR(s)	UR-AS-021
Command / action	jrc_as_variables 500 0.01 0.10 --alpha 2.0
Rationale	Confirms the script rejects an --alpha value outside (0, 1).
Pass criterion	Exit code ≠ 0.

TC-AS-VAR-011 Bypass protection

Tests UR(s)	UR-AS-022
Command / action	Rscript repos/as/R/jrc_as_variables.R 500 0.01 0.10 (RENV_PATHS_ROOT unset)
Rationale	Confirms that running the R script directly without the validated wrapper exits non-zero and mentions RENV_PATHS_ROOT in the error output.
Pass criterion	Exit code ≠ 0. 'RENV_PATHS_ROOT' appears in combined stdout/stderr.

6. Test Cases — jrc_as_oc_curve

TC-AS-OCC-001 Happy path — exit 0, Pa values in output

Tests UR(s)	UR-AS-023
Command / action	jrc_as_oc_curve 32 1
Rationale	Confirms valid n and c produce exit 0 and Pa values.
Pass criterion	Exit code = 0. Output contains 'Pa' and at least one numeric Pa value.

TC-AS-OCC-002 Pa at p=0.001 is > 0.99

Tests UR(s)	UR-AS-024
Command / action	jrc_as_oc_curve 32 1
Rationale	Confirms the plan (n=32, c=1) has very high acceptance probability at very low defect rates.
Pass criterion	Pa value in the first data row (p=0.001) extracted from output is > 0.99.

TC-AS-OCC-003 Pa decreases as p increases

Tests UR(s)	UR-AS-025
Command / action	jrc_as_oc_curve 32 1
Rationale	Confirms the OC curve is monotonically decreasing in p.
Pass criterion	Pa value at first p in the table > Pa value at last p in the table.

TC-AS-OCC-004 --lot-size accepted

Tests UR(s)	UR-AS-026
Command / action	jrc_as_oc_curve 32 1 --lot-size 200
Rationale	Confirms the --lot-size flag is accepted without error.

Pass criterion	Exit code = 0.

TC-AS-OCC-005 PNG saved to ~/Downloads/

Tests UR(s)	UR-AS-027
Command / action	jrc_as_oc_curve 32 1
Rationale	Confirms a PNG file is created in ~/Downloads/ following the script run.
Pass criterion	A file matching *_jrc_as_oc_curve.png exists in ~/Downloads/ with mtime ≥ test start time.

TC-AS-OCC-006 --aql and --rql accepted, AQL in output

Tests UR(s)	UR-AS-028
Command / action	jrc_as_oc_curve 32 1 --aql 0.01 --rql 0.10
Rationale	Confirms --aql and --rql flags are accepted and reflected in output.
Pass criterion	Exit code = 0. Output contains 'AQL'.

TC-AS-OCC-007 No arguments — non-zero exit

Tests UR(s)	UR-AS-029
Command / action	jrc_as_oc_curve (no arguments)
Rationale	Confirms the script exits non-zero when called with no arguments.
Pass criterion	Exit code ≠ 0.

TC-AS-OCC-008 c ≥ n — non-zero exit

Tests UR(s)	UR-AS-030
Command / action	jrc_as_oc_curve 10 10
Rationale	Confirms the script rejects c ≥ n (acceptance number must be less than sample size).
Pass criterion	Exit code ≠ 0.

TC-AS-OCC-009 n ≤ 0 — non-zero exit

Tests UR(s)	UR-AS-031
Command / action	jrc_as_oc_curve 0 0
Rationale	Confirms the script rejects n ≤ 0.
Pass criterion	Exit code ≠ 0.

TC-AS-OCC-010 Bypass protection

Tests UR(s)	UR-AS-032
Command / action	Rscript repos/as/R/jrc_as_oc_curve.R 32 1 (RENV_PATHS_ROOT unset)
Rationale	Confirms that running the R script directly without the validated wrapper exits non-zero and mentions RENV_PATHS_ROOT in the error output.
Pass criterion	Exit code ≠ 0. 'RENV_PATHS_ROOT' appears in combined stdout/stderr.

7. Test Cases — jrc_as_evaluate

TC-AS-EVAL-001 Attributes ACCEPT verdict

Tests UR(s)	UR-AS-033
Command / action	jrc_as_evaluate oq/data/attr_accept_lot.csv --type attributes --c 2
Rationale	Confirms ACCEPT verdict for a lot with 1 defective against c=2. The test lot (attr_accept_lot.csv) contains 32 units with 1 defective.
Pass criterion	Exit code = 0. Output contains 'ACCEPT'.

TC-AS-EVAL-002 Attributes REJECT verdict

Tests UR(s)	UR-AS-034
Command / action	jrc_as_evaluate oq/data/attr_reject_lot.csv --type attributes --c 2
Rationale	Confirms REJECT verdict for a lot with 5 defectives against c=2. The test lot (attr_reject_lot.csv) contains 32 units with 5 defectives.
Pass criterion	Exit code = 0. Output contains 'REJECT'.

TC-AS-EVAL-003 Variables ACCEPT verdict

Tests UR(s)	UR-AS-035
Command / action	jrc_as_evaluate oq/data/var_accept_lot.csv --type variables --k 1.45 --lsl 9.5
Rationale	Confirms ACCEPT verdict for a lot where Q_L >> k. The test lot (var_accept_lot.csv) has mean ≈10.45 and SD ≈0.051, giving Q_L ≈18.5.
Pass criterion	Exit code = 0. Output contains 'ACCEPT'.

TC-AS-EVAL-004 Variables REJECT verdict

Tests UR(s)	UR-AS-036
Command / action	jrc_as_evaluate oq/data/var_reject_lot.csv --type variables --k 1.45 --lsl 9.5
Rationale	Confirms REJECT verdict for a lot where Q_L < k. The test lot (var_reject_lot.csv) has mean ≈9.6 and SD ≈0.103, giving Q_L ≈0.975.
Pass criterion	Exit code = 0. Output contains 'REJECT'.

TC-AS-EVAL-005 Variables mode PNG saved

Tests UR(s)	UR-AS-037
Command / action	jrc_as_evaluate oq/data/var_accept_lot.csv --type variables --k 1.45 --lsl 9.5
Rationale	Confirms a PNG is created in ~/Downloads/ after a variables mode run.
Pass criterion	A file matching *_jrc_as_evaluate.png exists in ~/Downloads/ with mtime ≥ test start time.

TC-AS-EVAL-006 Attributes mode PNG saved

Tests UR(s)	UR-AS-038
Command / action	jrc_as_evaluate oq/data/attr_accept_lot.csv --type attributes --c 2
Rationale	Confirms a PNG is created in ~/Downloads/ after an attributes mode run.
Pass criterion	A file matching *_jrc_as_evaluate.png exists in ~/Downloads/ with mtime ≥ test start time.

TC-AS-EVAL-007 Missing --type — non-zero exit

Tests UR(s)	UR-AS-039
Command / action	jrc_as_evaluate oq/data/attr_accept_lot.csv --c 2
Rationale	Confirms the script exits non-zero when --type is omitted.
Pass criterion	Exit code ≠ 0.

TC-AS-EVAL-008 Attributes mode missing --c — non-zero exit

Tests UR(s)	UR-AS-040
Command / action	jrc_as_evaluate oq/data/attr_accept_lot.csv --type attributes
Rationale	Confirms the script exits non-zero when --type attributes is specified without --c.
Pass criterion	Exit code ≠ 0.

TC-AS-EVAL-009 Variables mode missing --k — non-zero exit

Tests UR(s)	UR-AS-041
Command / action	jrc_as_evaluate oq/data/var_accept_lot.csv --type variables --lsl 9.5
Rationale	Confirms the script exits non-zero when --type variables is specified without --k.
Pass criterion	Exit code ≠ 0.

TC-AS-EVAL-010 Attributes: missing 'result' column

Tests UR(s)	UR-AS-042
Command / action	jrc_as_evaluate oq/data/attr_missing_result.csv --type attributes --c 2
Rationale	Confirms the script exits non-zero and reports the missing 'result' column when an attributes CSV has the wrong column name.
Pass criterion	Exit code ≠ 0. 'result' appears in combined stdout/stderr.

TC-AS-EVAL-011 Variables: missing 'value' column

Tests UR(s)	UR-AS-043
Command / action	jrc_as_evaluate oq/data/var_missing_value.csv --type variables --k 1.45 --lsl 9.5
Rationale	Confirms the script exits non-zero and reports the missing 'value' column when a variables CSV has the wrong column name.
Pass criterion	Exit code ≠ 0. 'value' appears in combined stdout/stderr.

TC-AS-EVAL-012 Bypass protection

Tests UR(s)	UR-AS-044
Command / action	Rscript repos/as/R/jrc_as_evaluate.R oq/data/attr_accept_lot.csv ... (RENV_PATHS_ROOT unset)
Rationale	Confirms that running the R script directly without the validated wrapper exits non-zero and mentions RENV_PATHS_ROOT in the error output.
Pass criterion	Exit code ≠ 0. 'RENV_PATHS_ROOT' appears in combined stdout/stderr.

8. Requirements Traceability Matrix

Maps each test case to the user requirement(s) it covers.

Test Case ID	User Requirement(s)
TC-AS-ATTR-001	UR-AS-001
TC-AS-ATTR-002	UR-AS-002
TC-AS-ATTR-003	UR-AS-003
TC-AS-ATTR-004	UR-AS-004
TC-AS-ATTR-005	UR-AS-005
TC-AS-ATTR-006	UR-AS-006
TC-AS-ATTR-007	UR-AS-007
TC-AS-ATTR-008	UR-AS-008
TC-AS-ATTR-009	UR-AS-009
TC-AS-ATTR-010	UR-AS-010
TC-AS-ATTR-011	UR-AS-011
TC-AS-VAR-001	UR-AS-012
TC-AS-VAR-002	UR-AS-013
TC-AS-VAR-003	UR-AS-014
TC-AS-VAR-004	UR-AS-015
TC-AS-VAR-005	UR-AS-016
TC-AS-VAR-006	UR-AS-017
TC-AS-VAR-007	UR-AS-018
TC-AS-VAR-008	UR-AS-019
TC-AS-VAR-009	UR-AS-020
TC-AS-VAR-010	UR-AS-021
TC-AS-VAR-011	UR-AS-022
TC-AS-OCC-001	UR-AS-023
TC-AS-OCC-002	UR-AS-024
TC-AS-OCC-003	UR-AS-025
TC-AS-OCC-004	UR-AS-026
TC-AS-OCC-005	UR-AS-027

TC-AS-OCC-006	UR-AS-028
TC-AS-OCC-007	UR-AS-029
TC-AS-OCC-008	UR-AS-030
TC-AS-OCC-009	UR-AS-031
TC-AS-OCC-010	UR-AS-032
TC-AS-EVAL-001	UR-AS-033
TC-AS-EVAL-002	UR-AS-034
TC-AS-EVAL-003	UR-AS-035
TC-AS-EVAL-004	UR-AS-036
TC-AS-EVAL-005	UR-AS-037
TC-AS-EVAL-006	UR-AS-038
TC-AS-EVAL-007	UR-AS-039
TC-AS-EVAL-008	UR-AS-040
TC-AS-EVAL-009	UR-AS-041
TC-AS-EVAL-010	UR-AS-042
TC-AS-EVAL-011	UR-AS-043
TC-AS-EVAL-012	UR-AS-044